

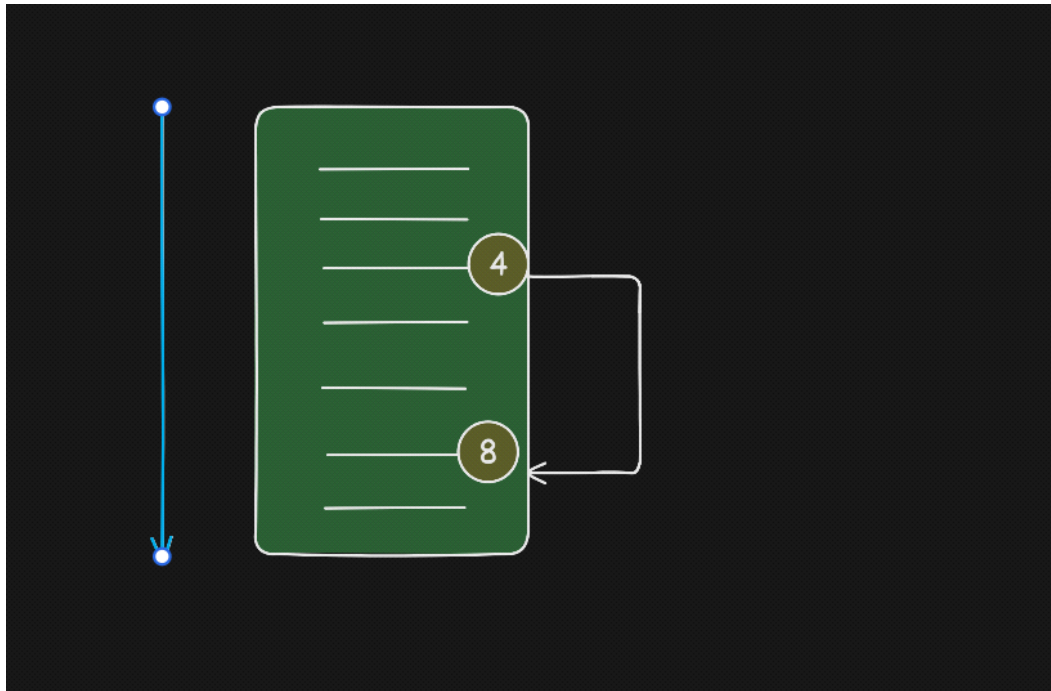
Control Statements

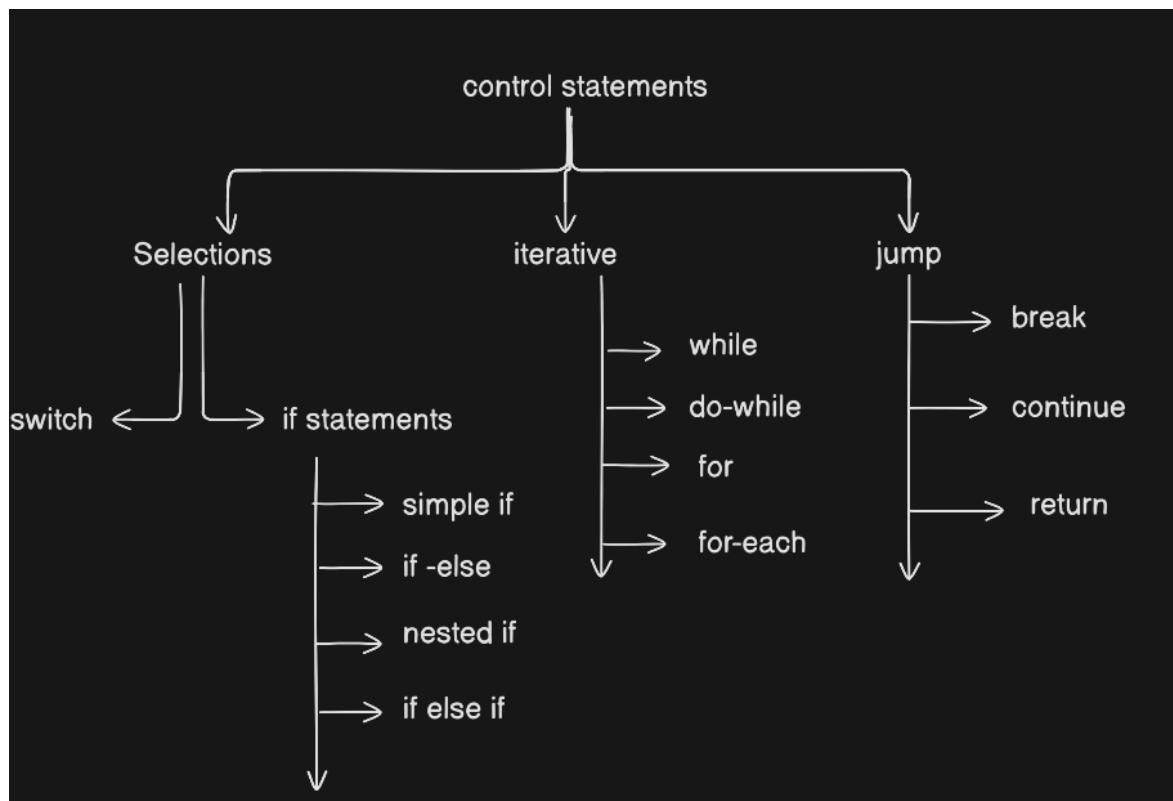
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Till Now, students have learned:

1. Data types
2. Variables
3. Operators

Now we have to learn how java make decision:





What are control statements ?

Control statements are used to control the flow of execution .

Control statement = statement that controls which code should execute.

Note: normally java executes code from top to bottom.

```
1 class ControlStatements
2 {
3     public static void main(String[] args)
4     {
5         System.out.println("Line 1");
6         System.out.println("Line 2");
7         System.out.println("Line 3");
8         System.out.println("Line 4");
9         System.out.println("Line 5");
10    }
11 }
12 }
```

```
D:\Spark6.0>java ControlStatements
Line 1
Line 2
Line 3
Line 4
Line 5

D:\Spark6.0>
```

But sometime we want code to execute only when a condition is true, for that we use decision making statements.

what is condition ?

Eg:

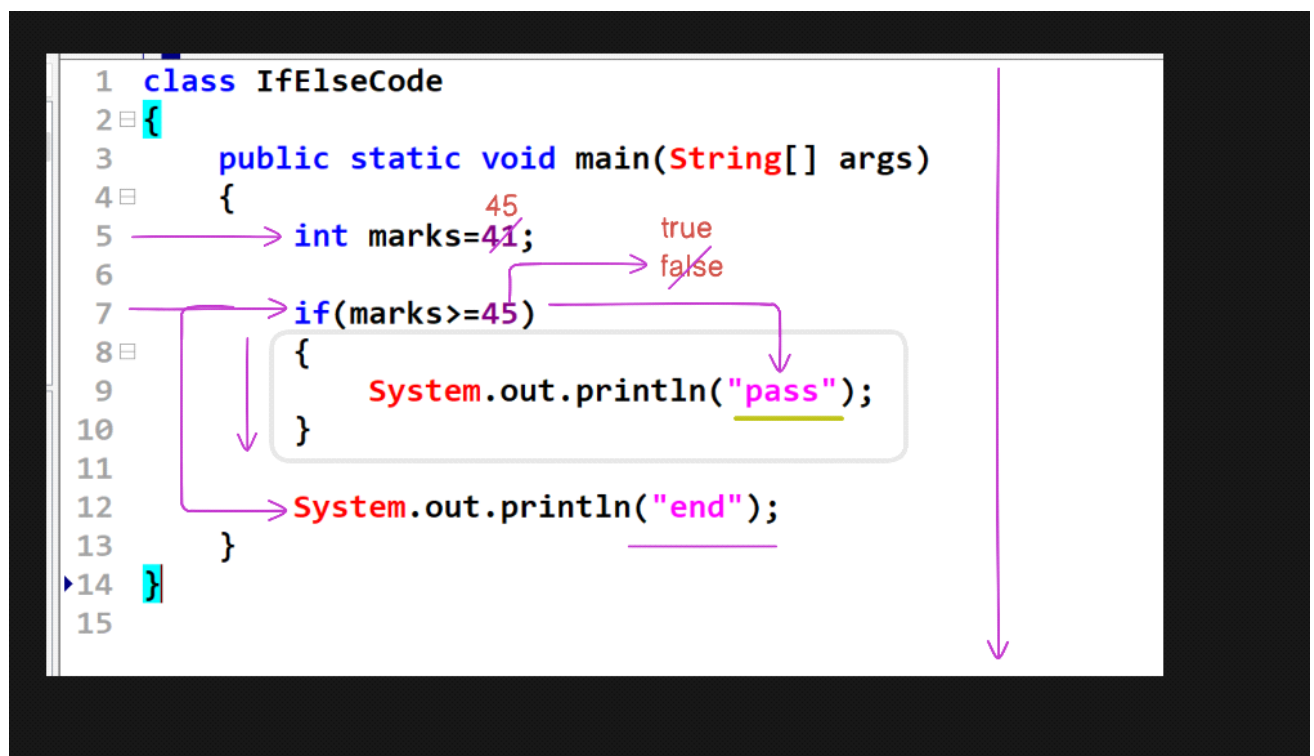
Age>=18

Marks>=45

Pas==123

Number %2==0

How if works:



Else-if

We use else-if when we have more then two possibilities.

```

1 class IfElseCode02
2 {
3     public static void main(String[] args)
4     {
5         int marks = 62;
6         if(marks >= 90)
7         {
8             System.out.println("Grade A+");
9         }
10        else if(marks >= 75)
11        {
12            System.out.println("Grade A");
13        }
14        else if(marks >= 60)
15        {
16            System.out.println("Grade B");
17        }
18        else if(marks >= 40)
19        {
20            System.out.println("Grade C");
21        }
22        else
23        {
24            System.out.println("Fail");
25        }
26    }
27 }
28 }

```

Annotations in the image:

- An arrow points from the value `62` in `int marks = 62;` to the first `if` condition `if(marks >= 90)`.
- An arrow points from the text `true` to the `System.out.println("Grade A+");` statement.
- An arrow points from the text `true` to the `System.out.println("Grade A");` statement.
- An arrow points from the text `if all blocks are not executed then this will work` to the `else` block.

Nested If

Nested if means if inside another if

Use nested if when second condition should be checked only after first condition is true

Switch

Switch is used when we compare one value to multiple fixed option

Real life example:

Days names

Grade

Month

Why we need it ?

If
If else
Else-if
Switch

Note:

Condition help java make decision

Loops help java repeating instruction

✓ Print hello world 10 --> ok will write 10 time

Print hello world 100 --> ok will write 100 times ✓

Print hello world 1000 ---> ok will write 1000 time ✗

Print hello world 100000 --> ok will not write ✗

System.out.println("Hello World!");

What is loop ?

loop is a terminology which help us to

1. Write less code
2. Less repetition
3. Easier to write

3 important parts of loop

-
1. Initialization --->
 2. Condition
 3. Update

Type of loops

-
1. While loop
 2. Do-while
 3. For loop

```

6      //initialization
7      int i=1; ✓
8
9      //condition
10     while(i<=10)
11     {
12         System.out.println("Hello World!");
13         i++; //i=i+1;
14     }
15
16
17 }

```

2. Do-while loop:

Intialization

Do{

}while()

Real world us of do-while

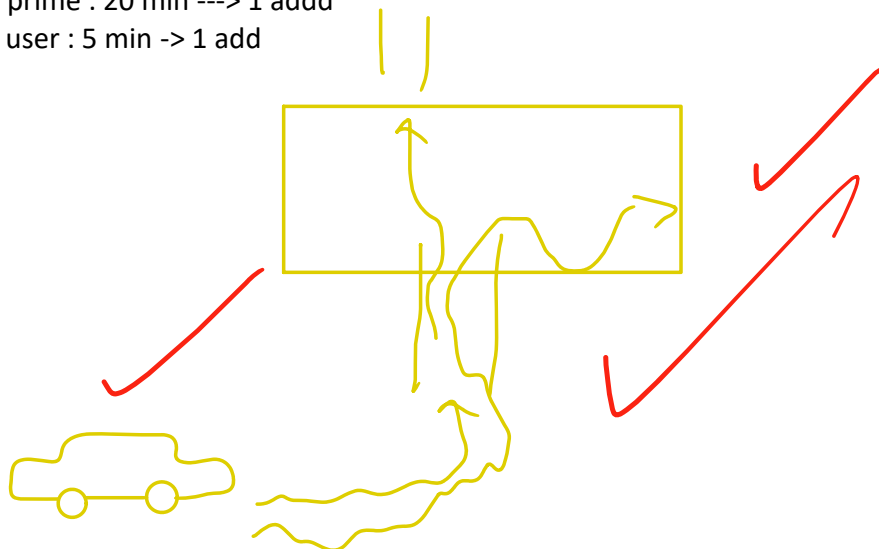
Example:

Normal: whey protein --- 100 -- do while

Gold user : whey protein - 1 -- do while

Amazon prime : 20 min ---> 1 addd

Normal user : 5 min -> 1 add



For loop

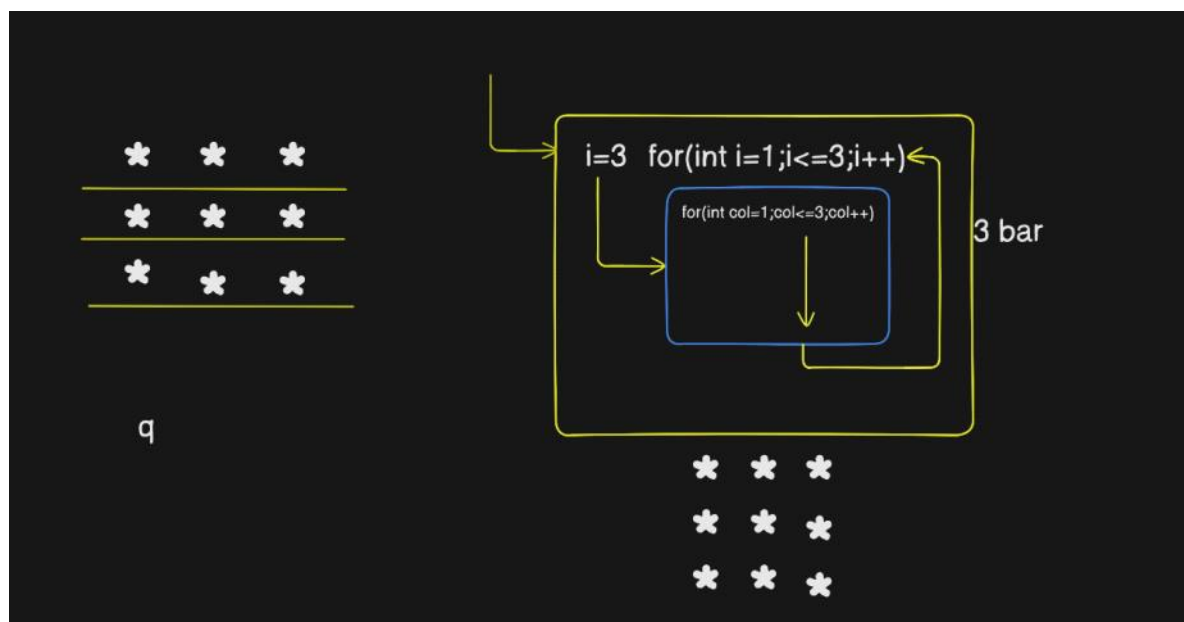
Use for loop when initialization, condition and update are known

Difference between loops

While	Do while	For
Checks condition first	Execute first	Check condition first
May run zero times	Run at least once	May run zero times
Good when repetition are uncertain	Good for menu/input entry, role based activates	Good when count is known
Initialization and condition are separate	Initialization and condition are separate	Initialization and condition are together

Nested Loops

A loop inside loop:



Ctrl + c

Break : 11:03-----> 11:25